

DATA ANALYSIS USING PYTHON-Internship

Task 2 (Easy) - Content Type Analysis Dashboard

#Step-1.

```
[8]:  
  
# TASK 2
```

```
[38]: import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt  
import seaborn as sns
```

```
[10]: # step 1 : Load the clean data set  
df=pd.read_csv("cleaned_netflix_titles.csv")  
df.head()
```

	show_id	type	title	director	country	date_added	release_year	rating	duration	listed_in
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	United States	2021-09-25	2020	PG-13	90 min	Documentaries
1	s3	Tv Show	Ganglands	Julien Leclercq	France	2021-09-24	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...
2	s6	Tv Show	Midnight Mass	Mike Flanagan	United States	2021-09-24	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries
3	s14	Movie	Confessions of an Invisible Girl	Bruno Garotti	Brazil	2021-09-22	2021	TV-PG	91 min	Children & Family Movies, Comedies
4	s8	Movie	Sankofa	Haile Gerima	United States	2021-09-24	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies

#Step-2.

```
# step 2: Calculate the total number of Movies and TV Show
```

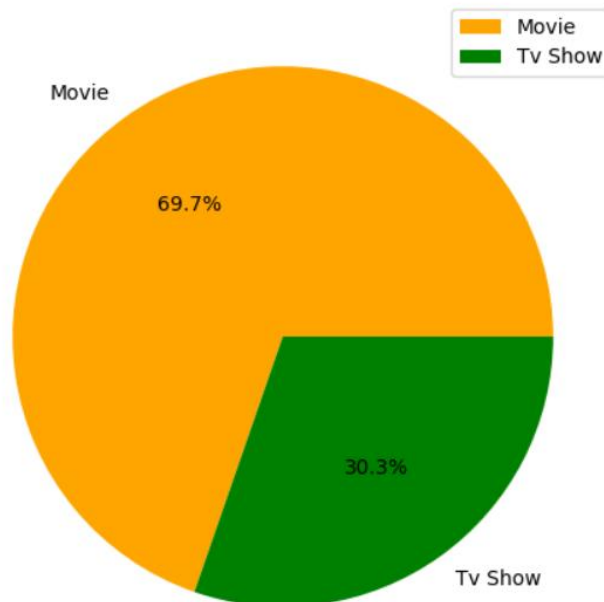
```
content_counts = df['type'].value_counts()  
print("The total Movie and TV Shows Count are:",  
      content_counts)
```

```
The total Movie and TV Shows Count are: type  
Movie      6126  
Tv Show    2664  
Name: count, dtype: int64
```

#step-3.

```
# step 3 :Create visualizations for content distribution
```

```
plt.figure(figsize=(8, 6))
plt.pie(content_counts, labels=content_counts.index, autopct='%1.1f%%', colors=['orange', 'green'])
plt.legend()
plt.show()
```



#Step-4.

```
# step 4: : Compare content proportions.
```

```
# 1. Get raw counts and calculate percentages simply
```

```
raw_counts = df['type'].value_counts()
```

```
proportions = df['type'].value_counts(normalize=True) * 100
```

```
print("      NETFLIX CONTENT TYPE DISTRIBUTION      ")
```

```
# Loop through categories (Movie and TV Show) simply
```

```
for category in raw_counts.index:
```

```
    count = raw_counts[category]
```

```
    percentage = proportions[category]
```

```
    print(f"• {category}: {count} titles ({percentage:.2f}%")
```

NETFLIX CONTENT TYPE DISTRIBUTION

- Movie: 6126 titles (69.69%)
- Tv Show: 2664 titles (30.31%)

Step-5: Summarize key findings.

Capital Allocation & Inventory volumetric : The numerical dominance of Movies (69.69% of total catalog volume) vs. TV Shows (30.31%) highlights a specific content-acquisition strategy.

- > Movies require a lower initial long-term operational commitment per title compared to multi-season TV series.
- > Historically, stocking the library with feature films allowed Netflix to rapidly scale its total library volume, creating a massive storefront of choice for subscriber acquisition across diverse demographics.